

GARP Risk and AI Exam Prep

100+ Practice Questions with Theory Slides

Unofficial Study Aid

Based on the official GARP Risk and AI curriculum

October 3, 2026

Outline

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- 3 Module 3: Risk and Risk Factors
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Module 1 — Question 1

Question

A bank uses AI models for detecting fraudulent transactions. While the bank's models are highly effective at identifying potential fraud, the risk management team is concerned about the inscrutability of the models. Which of the following statements correctly describes inscrutability in AI models?

- ☒ Inscrutability is the inability to understand the internal mechanisms of a model.
- ☐ Inscrutability is the inability to determine the accuracy of a model's prediction.
- ☐ Inscrutability is the inability to explain a model's predictions.
- ☐ Inscrutability is the inability to generalize a model to out-of-sample data.

Correct Answer: A

Explanation

Inscrutability refers to the inability to understand the internal mechanisms of a model — how it arrives at its outputs. It is distinct from accuracy, explainability, and generalizability.

Theory — Learn This for the Exam

Inscrutability = inability to understand the *internal mechanisms* of a model.

Explainability = ability to explain *why* a model made a specific decision.

Interpretability = ability to understand the *internal workings* of a model.

Key distinctions:

- A model can be accurate but inscrutable (e.g., deep neural networks).
- A model can be interpretable but not explainable in human terms (e.g., complex rule sets).
- Inscrutability is a **model risk** because it hinders validation, audit, and regulatory compliance.
- GARP exam frequently tests the difference between these three terms.

Module 1 — Question 2

Question

In March 1997, Deep Blue, an AI model developed by IBM, defeated the World Chess Champion Garry Kasparov. This event was considered a success for classical AI (also known as Good Old-Fashioned AI) development. Which of the following best describes a characteristic of classical AI?

- ☐ Classical AI can optimize its performance by using the gradient descent method.
- ☐ Classical AI cannot process structured data like time series data.
- ☒ Classical AI typically relies on pre-defined rules within simple contexts.
- ☐ Classical AI can easily scale to large, complex problems.

Correct Answer: C

Explanation

Classical AI (GOFAI) relies on pre-defined rules and logic within simple, well-defined contexts. It does not use gradient descent (that is deep learning) and struggles to scale to complex problems.

Module 1 — Theory: Classical AI vs. Deep Learning

Theory — Learn This for the Exam

Classical AI (GOFAI — Good Old-Fashioned AI):

- Rule-based, symbolic logic
- Uses search algorithms (A*, minimax, lookahead)
- Works well in *simple, well-defined* contexts (chess, tic-tac-toe)
- Cannot scale to large, complex, unstructured problems
- Does **not** learn from data using gradient descent

Deep Learning:

- Neural networks with many layers
- Learns from data via backpropagation and gradient descent
- Uses **self-training** (learning by competing against itself)
- Scales to complex problems (image recognition, NLP, generative AI)
- Can be inscrutable (black-box)

Exam tip: If a question mentions “self-training” or “gradient descent,” it is deep learning. If it mentions

Module 1 — Question 3

Question

Which of the following best describes a societal risk scenario regarding the impact of AI?

- ☐ A AI systems employed by a mid-size trading company makes errors resulting in losses for clients.
- ☒ B AI systems employed by a large social media company spread misinformation that influences public opinion.
- ☐ C AI systems used by a local manufacturer for hiring result in poor hiring decisions.
- ☐ D AI systems used by a local hospital access personal information resulting in a loss of privacy.

Correct Answer: B

Explanation

A societal risk affects society at large. Misinformation spread by a large social media company influencing public opinion is a societal risk. The other options describe organizational or individual risks.

Theory — Learn This for the Exam

Three levels of AI risk:

- **Individual risk:** harm to a specific person (privacy breach, wrong hiring decision, wrong medical diagnosis).
- **Organizational risk:** harm to a company (financial loss, reputational damage, regulatory fines).
- **Societal risk:** harm to society at large (misinformation, inequality, erosion of trust, democratic impact).

Examples:

- Hospital accessing personal info → individual privacy risk
- Trading company loses money → organizational risk
- Social media spreads misinformation → societal risk
- Hiring algorithm discriminates → individual + organizational risk

Exam tip: Read the scale of impact. “Society,” “public opinion,” “democracy” = societal.

Module 1 — Question 4

Question

One of the key breakthroughs of AI and machine learning is the development of deep learning. Which of the following is a key differentiating feature of deep learning over classical AI?

- ☐ A* ("A star") algorithm
- ☐ Recursive adversarial search
- ☐ Lookahead strategy
- ☐ Self-training

Correct Answer: D

Explanation

Self-training is an important aspect of deep learning, where the machine learning program learns by competing against itself. A*, recursive adversarial search, and lookahead are classical AI techniques.

Module 1 — Theory: Key Breakthroughs in AI

Theory — Learn This for the Exam

Timeline of AI breakthroughs:

- **1950s–1980s:** Classical AI (GOFAI) — logic, search, expert systems
- **1997:** Deep Blue beats Kasparov — classical AI success
- **2010s:** Deep learning revolution (AlexNet, AlphaGo)
- **2017:** Transformers introduced (“Attention Is All You Need”)
- **2020s:** Large Language Models (GPT, Claude, Gemini)

Key differentiating features of deep learning:

- Self-training (learning by competing against itself)
- Backpropagation + gradient descent
- Automatic feature extraction (no manual feature engineering)
- Handles unstructured data (images, text, audio)
- Inscrutable (black-box)

Module 1 — Question 5

Question

To better tailor its products and advisory services, a private wealth advisor surveyed potential customers about their investment preferences. One set of survey questions asked customers to select the types of investments (stocks, bonds, mutual funds, ETFs, etc.) they held within specific industry sectors. What would be the most appropriate way to classify the results from these survey questions?

- ☐ Ordinal data
- ☒ Nominal data
- ☐ Interval data
- ☐ Ratio data

Correct Answer: B

Explanation

The data are categorical with no inherent order — stocks, bonds, mutual funds, and ETFs are distinct categories without ranking. This is nominal data.

Module 1 — Theory: Data Types (Nominal, Ordinal, Interval, Ratio)

Theory — Learn This for the Exam

Four levels of measurement:

- **Nominal:** Categories with *no order* (stocks, bonds, ETFs; red, blue, green)
- **Ordinal:** Categories with *order but no equal intervals* (education level, Likert scale)
- **Interval:** Numeric with *equal intervals but no true zero* (temperature in Celsius, IQ scores)
- **Ratio:** Numeric with *equal intervals and a true zero* (income, age, height, weight)

Exam tip:

- “Types of” or “categories of” = nominal
- “Ranked” or “ordered” = ordinal
- “Temperature” or “score” = interval
- “Amount” or “count” = ratio

Module 1 — Question 6

Question

As part of cleaning and preparing a data set, an analyst is considering different data scaling techniques. Which of the following statements is correct regarding data scaling?

- ☐ Standardization creates a distribution bounded by 0 and 1.
- ☐ Standardization transforms a highly skewed distribution into a standard normal distribution.
- ☐ Normalization results in a data set bounded by -1 and 1.
- ☐ Normalization preserves the correlations between data points.

Correct Answer: D

Explanation

Standardization creates a distribution with mean 0 and standard deviation 1 (not bounded by 0 and 1). Normalization creates a dataset bounded by 0 and 1. Normalization preserves correlations between data points.

Module 1 — Theory: Standardization vs. Normalization

Theory — Learn This for the Exam

Standardization (Z-score):

- Formula: $z = (x - \mu) / \sigma$
- Result: mean = 0, standard deviation = 1
- **Not** bounded by 0 and 1
- Does **not** change the shape of the distribution
- Preserves correlations

Normalization (Min-Max):

- Formula: $x' = (x - \min) / (\max - \min)$
- Result: bounded by 0 and 1
- Preserves correlations between data points
- Sensitive to outliers (because min/max are affected by outliers)

Exam tip: “Bounded by 0 and 1” = normalization. “Mean 0, SD 1” = standardization. “Preserves correlations” = both, but normalization is the classic answer.

Module 1 — Question 7

Question

An analyst is researching the relationship between age and annual salary. Based on a review of published research, the analyst suspects that a polynomial regression is the best model to use. To determine the best form of the model, the analyst employs the forward stepwise regression technique. Based on the results of the first three steps shown below, which model should the analyst select?

Step	Model	AIC
0	No predictors	3202.69
1	Age	3193.36
1	Age2	3203.33
1	Age3	3204.66
1	Age4	3203.63
1	Age5	3202.20
2	Age + Age2	3125.50
2	Age + Age3	3123.20
2	Age + Age4	3126.69
2	Age + Age5	3132.87
3	Age + Age3 + Age2	3125.19

Module 1 — Theory: Stepwise Regression and AIC

Theory — Learn This for the Exam

Stepwise regression: automated feature selection method.

- **Forward:** start with no predictors, add one at a time
- **Backward:** start with all predictors, remove one at a time
- **Bidirectional:** combine forward and backward

AIC (Akaike Information Criterion):

- Measures model fit penalized by complexity
- **Lower AIC = better model**
- $AIC = 2k - 2\ln(L)$, where k = number of parameters, L = likelihood

How to use in stepwise:

- At each step, add/remove the variable that reduces AIC the most
- Stop when no further reduction is possible

Exam tip: Lower AIC is always better. If adding a variable increases AIC, stop.

Question

Which of the following is a key characteristic of machine learning compared to classical statistics?

- ☐ Machine learning focuses on inference and parameter estimation.
- ☒ Machine learning emphasizes prediction accuracy over model interpretability.
- ☐ Machine learning requires strict assumptions about data distribution.
- ☐ Machine learning is always more interpretable than classical statistics.

Correct Answer: B

Explanation

Machine learning prioritizes predictive performance over inference and interpretability. Classical statistics focuses on inference, parameter estimation, and assumptions about data distribution.

Theory — Learn This for the Exam

Classical statistics:

- Goal: inference (understand relationships, test hypotheses)
- Emphasizes parameter estimation and confidence intervals
- Requires strict assumptions (normality, independence, linearity)
- More interpretable (coefficients have meaning)

Machine learning:

- Goal: prediction (maximize accuracy on unseen data)
- Emphasizes generalization over inference
- Fewer distributional assumptions
- Often less interpretable (black-box)

Exam tip: “Inference” or “parameter estimation” = classical statistics. “Prediction” or “accuracy” = machine learning.

Module 1 — Question 9

Question

An analyst is preparing a dataset for a machine learning model. The dataset contains features with vastly different scales — age in years (20–80) and income in dollars (20,000–500,000). Which data preparation technique should the analyst apply?

- ☐ Remove the income feature
- ☒ Apply feature scaling
- ☐ Add more features
- ☐ Reduce the number of observations

Correct Answer: B

Explanation

Feature scaling (standardization or normalization) is essential when features have different scales, to prevent features with larger magnitudes from dominating the learning process.

Module 1 — Theory: Feature Scaling

Theory — Learn This for the Exam

Why scale features?

- Many algorithms (K-means, KNN, SVM, neural networks) use distance measures
- Features with larger scales dominate distance calculations
- Example: income (\$20k–\$500k) vs. age (20–80) — income would dominate

When to scale:

- Distance-based algorithms: K-means, KNN, SVM, PCA
- Gradient-based algorithms: neural networks, logistic regression
- **Not** needed for tree-based models (decision trees, random forests)

Methods:

- Standardization (Z-score): mean 0, SD 1
- Normalization (Min-Max): bounded 0–1

Question

What is the primary purpose of Principal Component Analysis (PCA) in machine learning?

- ☐ A To increase the number of features
- ☒ B To reduce dimensionality while preserving variance
- ☐ C To classify data into categories
- ☐ D To cluster data points

Correct Answer: B

Explanation

PCA is a dimensionality reduction technique that transforms data into a smaller set of uncorrelated components while preserving as much variance as possible.

Module 1 — Theory: Principal Component Analysis (PCA)

Theory — Learn This for the Exam

PCA = dimensionality reduction technique.

- Transforms original features into new uncorrelated components
- Components are ordered by variance explained
- First component captures the most variance
- Used to reduce features before clustering or regression

Benefits:

- Reduces overfitting
- Speeds up training
- Handles multicollinearity
- Enables visualization of high-dimensional data

Limitations:

- Components are linear combinations — harder to interpret
- Assumes linear relationships

Module 2 — Question 11

Question

How would you choose the number of clusters when using unsupervised learning?

- ☐ Always use $K = 2$
- ☒ Use a scree plot and look for the elbow
- ☐ Set K equal to the number of data points
- ☐ Use the highest possible K value

Correct Answer: B

Explanation

A scree plot charts inertia (WCSS) against the number of clusters. The “elbow” point where the rate of decrease sharply changes indicates the optimal K . Silhouette scores can also be used.

Theory — Learn This for the Exam

Scree plot (elbow method):

- Plot WCSS (inertia) vs. K
- Look for the “elbow” where the rate of decrease sharply changes
- Beyond the elbow, adding clusters gives diminishing returns

Silhouette score:

- For each point: $s = (b - a) / \max(a, b)$
- a = average distance to points in own cluster
- b = average distance to points in nearest other cluster
- Range: -1 to 1 ; higher is better
- Choose K with the highest average silhouette score

Rule of thumb (weak): $K \approx \sqrt{N/2}$

Module 2 — Question 12

Question

Explain the steps in using the K-means clustering algorithm.

- Specify K, scale features, select K random centroids, assign points, recalculate centroids, repeat until convergence
- Specify K, assign labels, train a classifier, evaluate accuracy
- Select K random points, remove outliers, apply PCA, cluster
- Normalize data, apply hierarchical clustering, cut the dendrogram

Correct Answer: A

Explanation

The K-means algorithm: (1) Specify K and choose a distance measure, (2) Scale features, (3) Select K random centroids, (4) Assign each point to nearest centroid, (5) Recalculate centroids, (6) Repeat steps 4–5 until centroids stabilize.

Theory — Learn This for the Exam

K-means algorithm (Lloyd's algorithm):

- 1 Specify K (number of clusters) and choose distance measure (Euclidean or Manhattan)
- 2 Scale features (standardization or normalization)
- 3 Select K random initial centroids
- 4 **Assignment step:** assign each point to nearest centroid
- 5 **Update step:** recalculate centroids as mean of assigned points
- 6 Repeat steps 4–5 until centroids stabilize (convergence)

Key properties:

- Guaranteed to converge (but may find local optimum)
- Sensitive to initial centroid positions
- Use K-means++ for better initialization
- Run multiple times and choose lowest WCSS

Module 2 — Question 13

Question

In practice, the K-means clustering algorithm is often carried out with several different initial values for the centroids. How would you choose between clusters that result from different initial choices for the centroids?

- ☐ A Select the solution with the highest WCSS
- ☒ B Select the solution with the lowest WCSS
- ☐ C Select the solution with the most clusters
- ☐ D Select the first solution found

Correct Answer: B

Explanation

Select the clustering solution with the lowest total inertia (WCSS), as this represents the best fit to the feature data.

Theory — Learn This for the Exam

Within-Cluster Sum of Squares (WCSS):

- Measures cluster *compactness*
- Sum of squared distances from each point to its centroid
- Lower WCSS = better fit
- Always decreases as K increases

Between-Cluster Sum of Squares (BCSS):

- Measures cluster *separation*
- Sum of squared distances from each cluster centroid to the overall centroid, weighted by cluster size
- Higher BCSS = better separation

Relationship: Minimizing WCSS implicitly maximizes BCSS. **Exam tip:** WCSS = compactness; BCSS = separation.

Question

How do you use the scree plot for choosing the number of K-means clusters?

- ☐ Look for the highest point
- ☐ Look for the lowest point
- ☒ Look for an “elbow” where the gradient changes from steep to almost flat
- ☐ Look for the point where WCSS equals zero

Correct Answer: C

Explanation

A scree plot has the number of clusters on the x-axis and WCSS on the y-axis. We look for an “elbow” where the gradient changes from steep to almost flat — that is the optimal K.

Theory — Learn This for the Exam

Scree plot anatomy:

- X-axis: number of clusters (K)
- Y-axis: WCSS (inertia)
- Curve always decreases as K increases
- Look for the “elbow” — the point of maximum curvature

How to identify the elbow:

- Before elbow: steep decrease (adding clusters helps a lot)
- After elbow: flat decrease (adding clusters helps little)
- Elbow = optimal trade-off between fit and complexity

Common pitfalls:

- Elbow may be ambiguous (multiple elbows)
- No elbow may be visible
- Use domain knowledge + silhouette score to confirm

Module 2 — Question 15

Question

What are the two types of hierarchical clustering? What are the advantages of hierarchical clustering?

- Divisive and agglomerative; advantages: no need to pre-define K, uncovers hierarchical relationships, dendrograms are interpretable
- Supervised and unsupervised; advantages: faster, more accurate
- Partitional and density-based; advantages: handles outliers, finds arbitrary shapes
- Single and complete linkage; advantages: simpler, more scalable

Correct Answer: A

Explanation

Hierarchical clustering can be divisive (top-down) or agglomerative (bottom-up). Advantages: does not require pre-defining K, uncovers hierarchical relationships, and dendrograms are straightforward to interpret.

Module 2 — Theory: Hierarchical Clustering

Theory — Learn This for the Exam

Two approaches:

- **Divisive (top-down):** start with all data in one cluster, split recursively
- **Agglomerative (bottom-up):** start with each point as its own cluster, merge recursively

Linkage criteria:

- **Single linkage:** distance = shortest distance between clusters
- **Complete linkage:** distance = greatest distance between clusters
- **Average linkage:** distance = average distance
- **Ward's method:** minimizes within-cluster variance

Advantages:

- No need to pre-define K
- Dendrogram shows all possible clusterings
- Uncovers nested relationships

Disadvantages: Computationally expensive for large N.

Module 2 — Question 16

Question

State whether the following is true or false: K-means with Euclidean distance can only be used when the clusters are approximately spherical.

- ☒ True
- ☐ False
- ☐ True only for large datasets
- ☐ True only for small datasets

Correct Answer: A

Explanation

True. K-means with Euclidean distance is based on linear distances and runs into problems when clusters are not approximately spherically shaped. Manhattan distance produces rhombus-shaped clusters.

Theory — Learn This for the Exam

K-means produces spherical clusters.

- Because it minimizes distance to centroids
- Euclidean distance → spherical clusters
- Manhattan distance → rhombus-shaped clusters

Problems with K-means:

- **Outliers:** can distort centroids or form their own cluster
- **Curse of dimensionality:** distances become similar in high dimensions
- **Non-spherical clusters:** fails on rings, moons, elongated shapes
- **Correlated features:** struggles when features are highly correlated

Solutions:

- Standardize data (min-max)
- Remove outliers
- Apply PCA to reduce dimensions

Module 2 — Question 17

Question

State whether the following is true or false: The within-cluster sum of squares will never rise when the number of clusters is increased in a K-means application.

- ☒ True
- ☐ False
- ☐ True only if K is even
- ☐ True only if K is odd

Correct Answer: A

Explanation

True. As K increases, the fit cannot get worse because each new cluster captures at least one data point better. This is analogous to residual sum of squares in regression.

Theory — Learn This for the Exam

WCSS is monotonically non-increasing in K.

- Adding a cluster can never increase WCSS
- At $K = N$, each point is its own cluster, $WCSS = 0$
- But $K = N$ is meaningless (overfitting)

Analogy:

- In regression, adding features never decreases R^2
- In K-means, adding clusters never increases WCSS

Implication:

- Cannot choose K by minimizing WCSS alone
- Must use elbow method, silhouette, or domain knowledge
- Balance fit vs. interpretability

Exam tip: WCSS always decreases or stays the same as K increases. Never rises.

Module 2 — Question 18

Question

What is a dendrogram and how would we interpret one?

- ☐ A bar chart showing feature importance
- ☒ A tree-like diagram showing hierarchical clustering steps; long vertical lines indicate important clusters
- ☐ A scatter plot of clusters
- ☐ A line chart of WCSS vs. K

Correct Answer: B

Explanation

A dendrogram is a pictorial representation of hierarchical clustering. Each bifurcation or combination shows a cluster being formed. Long vertical lines suggest the additional cluster has a considerable effect on model fit.

Theory — Learn This for the Exam

Dendrogram = tree-like diagram of hierarchical clustering.

- X-axis: data points (or clusters)
- Y-axis: distance (or dissimilarity)
- Each merge/split is shown as a horizontal line
- Height of vertical lines = distance at which merge occurs

Interpretation:

- Long vertical lines → important clusters (big effect on fit)
- Short vertical lines → minor clusters
- Cut the dendrogram horizontally to choose K
- Number of vertical lines crossed = number of clusters

Exam tip: Long vertical lines = significant cluster boundary. Cut horizontally where the gap is largest.

Module 2 — Question 19

Question

Suppose that we have the following data on three features for each of three banks, A, B, and C. Determine the centroid of a cluster comprising banks A, B, and C using the raw (unscaled) data.

Feature	Bank A	Bank B	Bank C
Number of customers (millions)	1.2	6.0	0.5
Total size of loan book (USD bn)	5	25	7
Number of branches	80	400	50

-  (2.567, 12.333, 176.667)
-  (2.567, 12.333, 176.667)
-  (2.567, 12.333, 176.667)
-  (2.567, 12.333, 176.667)

Correct Answer: A

Explanation

Theory — Learn This for the Exam

Centroid = mean of all points in a cluster.

- For each feature, take the average across all points in the cluster
- In K-means, centroids are recalculated each iteration
- Centroids do not have to correspond to actual data points

Formula:

$$\mu_j = \frac{1}{n_k} \sum_{i \in C_k} x_i$$

where C_k is the set of points in cluster k , n_k is the number of points. **Exam tip:** Centroid = average. Just add and divide by count for each feature.

Question

An analyst is given a set of data with two features. As part of the exploratory data analysis process, the analyst examines a scatter plot of the data. Which of the following best identifies an issue with the data that might pose problems for the K-means clustering algorithm?

- ☐ A Outliers in the data
- ☐ B High dimensionality of the data
- ☐ C Non-linearity of the relationships in the data
- ☒ D Non-spherical clusters in the data

Correct Answer: D

Explanation

K-means struggles with non-spherical clusters because it is based on distances from centroids and tends to produce spherical clusters.

Theory — Learn This for the Exam

Two pathological cases for K-means:

- **Concentric circles:** two rings, one inside the other. Centroids end up in the same place, algorithm cannot separate them.
- **Elongated clusters:** two ovals side by side. Points on the edge of one oval may be closer to the other cluster's centroid.

Why K-means fails:

- Assumes clusters are spherical and similar size
- Uses distance to centroid only
- Cannot capture complex shapes

Solutions:

- Kernel K-means (non-linear transformation)
- DBSCAN (density-based)
- Spectral clustering
- Gaussian Mixture Models

Question

What is the main advantage of DBSCAN over K-means?

- ☐ DBSCAN is faster
- ☒ DBSCAN can find clusters of arbitrary shapes and handle outliers
- ☐ DBSCAN requires fewer parameters
- ☐ DBSCAN always produces spherical clusters

Correct Answer: B

Explanation

DBSCAN defines clusters as dense regions separated by low-density regions. It can find arbitrary-shaped clusters and is robust to outliers (noise points).

Theory — Learn This for the Exam

DBSCAN = Density-Based Spatial Clustering of Applications with Noise.

- Defines clusters as *dense regions* separated by low-density regions
- Three types of points:
 - **Core point:** has at least minPts neighbors within radius ϵ
 - **Border point:** within ϵ of a core point but not a core point itself
 - **Noise point:** neither core nor border $\rightarrow$ outlier

Advantages:

- Finds arbitrary-shaped clusters (not just spherical)
- Robust to outliers (noise points are excluded)
- Does not require specifying K

Disadvantages:

- Struggles with varying densities
- Sensitive to ϵ and minPts

Question

What is the purpose of the silhouette score in clustering?

- ☐ To measure the accuracy of predictions
- ☒ To compare cluster cohesion and separation
- ☐ To count the number of clusters
- ☐ To visualize high-dimensional data

Correct Answer: B

Explanation

The silhouette score compares the average distance of a point to its own cluster (cohesion) with the average distance to the nearest other cluster (separation). Higher scores indicate better-defined clusters.

Theory — Learn This for the Exam

Silhouette score for point i :

$$s_i = \frac{b_i - a_i}{\max(b_i, a_i)}$$

- a_i = mean distance to points in own cluster (cohesion)
- b_i = mean distance to points in nearest other cluster (separation)
- Range: -1 to 1

Interpretation:

- $s \approx 1$: point is well inside its cluster
- $s \approx 0$: point is on the boundary between clusters
- $s < 0$: point may be mis-clustered

Overall silhouette score: average of all s_i . **Choose K with highest average silhouette score.**

Question

What is the curse of dimensionality in the context of K-means clustering?

- ☐ A The algorithm becomes too slow with many observations
- ☒ B Data points become sparse in high-dimensional space, making clustering difficult
- ☐ C The number of clusters increases exponentially
- ☐ D The centroids become unstable

Correct Answer: B

Explanation

As the number of features increases, data points become increasingly sparse in feature space. Distances between points and centroids become more similar, making it hard for K-means to identify clusters.

Theory — Learn This for the Exam

Curse of dimensionality: as features increase, data becomes sparse.

- Distances between points become more similar
- K-means cannot distinguish clusters
- Convergence slows down

Solutions:

- Remove irrelevant features
- Apply PCA or other dimensionality reduction
- Use cosine distance (does not grow with dimensions)
- Use algorithms designed for high dimensions

Exam tip: “Curse of dimensionality” = problems caused by too many features.

Module 2 — Question 24

Question

What is the difference between single linkage and complete linkage in hierarchical clustering?

- ☐ Single linkage uses the shortest distance between clusters; complete linkage uses the greatest distance
- ☐ Single linkage uses the greatest distance; complete linkage uses the shortest distance
- ☐ Single linkage is for agglomerative; complete linkage is for divisive
- ☐ Single linkage is for divisive; complete linkage is for agglomerative

Correct Answer: A

Explanation

Single linkage defines cluster distance as the shortest distance between any two points in different clusters. Complete linkage uses the greatest distance.

Theory — Learn This for the Exam

Linkage = how to measure distance between clusters.

- **Single linkage:** shortest distance between any two points
- **Complete linkage:** greatest distance between any two points
- **Average linkage:** average distance between all pairs
- **Ward's method:** minimizes within-cluster variance

Properties:

- Single linkage: can produce elongated clusters, sensitive to outliers
- Complete linkage: produces compact clusters, robust to outliers
- Average linkage: balance between single and complete

Exam tip: Single = shortest. Complete = greatest. Both apply to agglomerative and divisive.

Question

What is the purpose of K-means++ initialization?

- ☒ To speed up convergence by choosing centroids far apart
- ☐ To reduce the number of clusters
- ☐ To increase the WCSS
- ☐ To eliminate outliers

Correct Answer: A

Explanation

K-means++ selects initial centroids that are far apart, reducing the chance of poor clustering and helping the algorithm converge more quickly to a good solution.

Theory — Learn This for the Exam

Problem: Random initialization can lead to poor clustering.

- If centroids start close together, algorithm may converge to local optimum

K-means++ solution:

- 1 Choose first centroid randomly from data points
- 2 For each subsequent centroid, choose a point with probability proportional to its squared distance from the nearest existing centroid
- 3 This spreads centroids out

Benefits:

- Better clustering results
- Faster convergence
- Reduces chance of poor local optima

Exam tip: K-means++ = smarter initialization = centroids far apart.

Module 2 — Question 26

Question

A software engineer is using a facial recognition system. The engineer provides an input matrix X and a kernel matrix W . What is the correct feature map (F)?

$$X = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & 0 \\ 3 & 2 & 2 \end{pmatrix}, W = \begin{pmatrix} -1 & 0 \\ 1 & 1 \end{pmatrix}$$

☐ $F = \begin{pmatrix} 1 & -1 \\ 4 & 4 \end{pmatrix}$

☐ $F = \begin{pmatrix} 1 & 4 \\ -1 & 4 \end{pmatrix}$

☐ $F = \begin{pmatrix} 2 & 3 \\ 6 & 4 \end{pmatrix}$

☐ $F = \begin{pmatrix} 2 & 6 \\ 3 & 4 \end{pmatrix}$

Correct Answer: A

Theory — Learn This for the Exam

Convolution = element-wise multiplication + sum.

- Slide kernel W over input X
- At each position, multiply overlapping elements and sum
- Result is the feature map F

Example (2x2 kernel on 3x3 input):

- Output size = $(3 - 2 + 1) \times (3 - 2 + 1) = 2 \times 2$
- $F_{11} = X_{11}W_{11} + X_{12}W_{12} + X_{21}W_{21} + X_{22}W_{22}$
- $F_{12} = X_{12}W_{11} + X_{13}W_{12} + X_{22}W_{21} + X_{23}W_{22}$
- And so on...

Used in: CNNs for image recognition, facial recognition, feature extraction.

Module 2 — Question 27

Question

An analyst is using an ensemble technique to improve the performance of a decision tree prediction model. The analyst samples with replacement from the training data and creates multiple decision trees using all available features. The analyst then averages the results. Which technique is this?

- ☐ Adaptive boosting
- ☐ Random forest
- ☐ Gradient boosting
- ☐ Bootstrap aggregation

Correct Answer: D

Explanation

The question describes bootstrap aggregation (bagging). Random forest would use a random subset of features at each split, not all available features.

Theory — Learn This for the Exam

Ensemble = combine multiple models for better performance.

- **Bagging (Bootstrap Aggregation):** train models on bootstrap samples, average predictions. Uses *all* features.
- **Random Forest:** bagging + random feature subset at each split. Reduces correlation across trees.
- **Boosting:** train models sequentially, each focusing on previous errors.
 - **Adaptive Boosting (AdaBoost):** reweights misclassified points
 - **Gradient Boosting:** fits new model to residuals

Exam tip:

- “All features” + “average” = bagging
- “Random subset of features” = random forest
- “Sequential” + “correct errors” = boosting

Module 2 — Question 28

Question

A data scientist is working to improve the performance of a decision tree model. The data scientist thinks the model can be improved by implementing a random forest technique. Which of the following best describes an advantage of using a random forest?

- ☐ Random forest can capture previous misclassifications through self-learning
- ☐ Random forest can improve interpretability by linking information from previously grown trees
- ☐ Random forest can improve prediction accuracy by reducing correlation across trees
- ☐ Random forest can capture outliers by calculating Cook's distance

Correct Answer: C

Explanation

In random forest, some trees ignore some strong features and give other features a chance, resulting in less correlated forecasts and improved prediction accuracy.

Theory — Learn This for the Exam

Random Forest = bagging + random feature subset.

- Each tree is trained on a bootstrap sample
- At each split, consider only a random subset of features
- Final prediction = average (regression) or majority vote (classification)

Advantages:

- Reduces overfitting compared to single tree
- Reduces correlation across trees
- Handles high-dimensional data
- Robust to outliers
- Provides feature importance

Disadvantages:

- Less interpretable than single tree
- Computationally expensive

Module 2 — Question 29

Question

An asset manager uses an SVM model. The boundary line is $0.43x_1 + 2.55x_2 + 0.67 = 0$. For a stock with $x_1 = 0.8$ and $x_2 = -0.75$, which is the correct classification?

- ☐ Class 0, because the decision value is lower than 0
- ☐ Class 0, because the decision value is higher than 0
- ☐ Class 1, because the decision value is lower than 0
- ☐ Class 1, because the decision value is higher than 0

Correct Answer: A

Explanation

Substituting: $0.43(0.8) + 2.55(-0.75) + 0.67 = 0.344 - 1.9125 + 0.67 = -0.8985$. Since this is below zero, the stock belongs to Class 0.

Module 2 — Theory: Support Vector Machines (SVM)

Theory — Learn This for the Exam

SVM = finds the optimal hyperplane separating classes.

- Boundary: $w_1x_1 + w_2x_2 + b = 0$
- Decision value: $f(x) = w_1x_1 + w_2x_2 + b$
- If $f(x) > 0$: Class 1
- If $f(x) < 0$: Class 0

Key concepts:

- **Support vectors:** points closest to the boundary
- **Margin:** distance between boundary and support vectors
- **Kernel trick:** maps data to higher dimensions for non-linear separation

Extensions:

- Linear SVM (linear boundary)
- Polynomial kernel
- Radial Basis Function (RBF) kernel

Module 2 — Question 30

Question

An analyst trained four models to classify consumer loans. Based on the results, which model correctly identifies the highest percent of defaulted loans?

Model	Accuracy	Precision	Recall	F1
A (Logistic regression)	0.71	0.60	0.28	0.38
B (Unrestricted decision tree)	0.66	0.43	0.17	0.24
C (SVM)	0.63	0.44	0.51	0.47
D (K-nearest neighbors)	0.68	0.50	0.15	0.23

- ☐ Model A
- ☐ Model B
- ☐ Model C
- ☐ Model D

Correct Answer: C

Explanation

Theory — Learn This for the Exam

Confusion matrix:

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Metrics:

- **Accuracy** = $(TP + TN) / \text{Total}$
- **Precision** = $TP / (TP + FP)$ — how many predicted positives are correct
- **Recall (Sensitivity)** = $TP / (TP + FN)$ — how many actual positives are captured
- **F1** = $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$
- **Specificity** = $TN / (TN + FP)$

Exam tip: “Highest percent of defaulted loans” = highest recall.

Module 2 — Question 31

Question

A newly hired analyst has been asked to “tokenize” a text passage as part of data pre-processing for NLP. Which of the following best describes tokenization?

- ☐ Remove words that have no informational value
- ☒ Separate text sentences into words
- ☐ Convert each word into a numeric vector
- ☐ Replace words with similar meanings with one unique word

Correct Answer: B

Explanation

Tokenization is the process of separating text into individual words or tokens. Stop word removal, feature extraction, and lemmatization are different pre-processing steps.

Theory — Learn This for the Exam

NLP pre-processing steps:

- 1 **Tokenization:** split text into words/tokens
- 2 **Stop word removal:** remove words with no informational value (the, a, is)
- 3 **Lemmatization:** reduce words to base form (running → run)
- 4 **Stemming:** crude version of lemmatization (running → run)
- 5 **Feature extraction:** convert words to numeric vectors (TF-IDF, word embeddings)

Exam tip:

- “Separate into words” = tokenization
- “Remove common words” = stop word removal
- “Convert to numbers” = feature extraction
- “Replace with base form” = lemmatization

Question

What does the Continuous Bag of Words (CBoW) architecture represent?

- ☐ A recurrent neural network
- ☒ A “fill in the blank” technique that predicts a missing word
- ☐ A skip-gram model
- ☐ An auto-encoder

Correct Answer: B

Explanation

CBoW is a “fill in the blank” technique that proposes a probability-ordered list of words that would fill in the gap between words before and after a missing word.

Theory — Learn This for the Exam

Word2Vec = word embedding technique.

- Converts words to dense vectors
- Captures semantic relationships

Two architectures:

- **CBow (Continuous Bag of Words):**

- Predicts a missing word from context words
- “Fill in the blank”
- Multiple input neurons, one output neuron

- **Skip-gram:**

- Predicts context words from a target word
- One input neuron, multiple output neurons
- Works better with rare words

Exam tip: CBoW = many inputs, one output. Skip-gram = one input, many outputs.

Module 2 — Question 33

Question

An analyst is working on a text classification task with a small set of labeled articles and a larger set of unlabeled articles. The analyst plans to use a co-training approach. What is an advantage of co-training?

- Co-training provides different views of the data by letting the two models learn from each other
- Co-training has higher productivity since the technique employs parallel computing
- Co-training improves accuracy by decreasing the weights on unrelated features
- Co-training is more reliable since the data is not interchangeable between the two models

Correct Answer: A

Explanation

Co-training provides different views of the data by letting the two models learn from each other, improving performance with limited labeled data.

Theory — Learn This for the Exam

Semi-supervised learning = small labeled + large unlabeled data.

- Between supervised and unsupervised
- Uses unlabeled data to improve performance

Techniques:

- **Self-training:** train on labeled, predict unlabeled, add confident predictions to training set, repeat
- **Co-training:** split features into two views, train two models, each labels data for the other
- **Unsupervised pre-processing:** use clustering to initialize supervised model

Assumptions:

- **Smoothness:** nearby points have similar labels
- **Cluster:** points in same cluster have same label
- **Manifold:** data lies on a lower-dimensional manifold

Question

What is the primary purpose of regularization in machine learning?

- ☐ To increase model complexity
- ☒ To prevent overfitting by penalizing large coefficients
- ☐ To speed up training
- ☐ To increase the number of features

Correct Answer: B

Explanation

Regularization (Ridge, LASSO, Elastic Net) prevents overfitting by adding a penalty term for large coefficients, discouraging complex models.

Theory — Learn This for the Exam

Regularization = penalize complexity to prevent overfitting.

- **Ridge (L2):** adds $\lambda \sum \beta_j^2$ to loss
- **LASSO (L1):** adds $\lambda \sum |\beta_j|$ to loss
- **Elastic Net:** combines L1 and L2

Properties:

- Ridge: shrinks coefficients, keeps all features
- LASSO: can shrink coefficients to exactly zero (feature selection)
- Elastic Net: balance between Ridge and LASSO
- λ controls strength of regularization

Exam tip:

- “Feature selection” = LASSO
- “Shrink coefficients” = Ridge
- “Both” = Elastic Net

Question

What is the difference between Ridge regression and LASSO?

- ☐ Ridge uses L1 penalty; LASSO uses L2 penalty
- ☒ Ridge uses L2 penalty; LASSO uses L1 penalty
- ☐ Both use L1 penalty
- ☐ Both use L2 penalty

Correct Answer: B

Explanation

Ridge regression uses L2 penalty (sum of squared coefficients). LASSO uses L1 penalty (sum of absolute coefficients), which can shrink some coefficients to exactly zero, performing feature selection.

Theory — Learn This for the Exam

Ridge (L2):

$$\text{Loss} = \text{RSS} + \lambda \sum_{j=1}^p \beta_j^2$$

- Shrinks coefficients toward zero
- Never exactly zero
- Keeps all features
- Good when all features are useful

LASSO (L1):

$$\text{Loss} = \text{RSS} + \lambda \sum_{j=1}^p |\beta_j|$$

- Can shrink coefficients to exactly zero
- Performs feature selection

Question

What is the purpose of cross-validation?

- ☐ To increase the training set size
- ☒ To estimate model performance on unseen data
- ☐ To reduce the number of features
- ☐ To speed up training

Correct Answer: B

Explanation

Cross-validation splits data into folds, training on some and validating on others, to estimate how well the model generalizes to unseen data.

Theory — Learn This for the Exam

Cross-validation = estimate generalization performance.

- **K-fold:** split data into K folds, train on K-1, validate on 1, repeat K times
- **Stratified K-fold:** preserve class proportions in each fold
- **Leave-One-Out (LOO):** $K = N$
- **Time-series CV:** respect temporal order

Benefits:

- More reliable performance estimate
- Uses all data for both training and validation
- Helps detect overfitting

Exam tip: Cross-validation is for *model evaluation*, not training.

Module 2 — Question 37

Question

What is the bias-variance tradeoff?

- ☐ The tradeoff between model accuracy and training time
- ☒ The tradeoff between underfitting and overfitting
- ☐ The tradeoff between the number of features and observations
- ☐ The tradeoff between precision and recall

Correct Answer: B

Explanation

High bias leads to underfitting (model too simple). High variance leads to overfitting (model too complex). The goal is to find the balance that minimizes total error.

Theory — Learn This for the Exam

Total error = Bias² + Variance + Irreducible Error.

- **Bias:** error from overly simple assumptions (underfitting)
- **Variance:** error from sensitivity to training data (overfitting)

Tradeoff:

- Simple model: high bias, low variance
- Complex model: low bias, high variance
- Optimal: balance between the two

How to diagnose:

- High training error + high test error = underfitting (high bias)
- Low training error + high test error = overfitting (high variance)

Exam tip: Underfitting = high bias. Overfitting = high variance.

Question

What is the purpose of a confusion matrix?

- ☐ To visualize the decision boundary
- ☒ To summarize classification performance with true/false positives and negatives
- ☐ To plot the ROC curve
- ☐ To cluster data points

Correct Answer: B

Explanation

A confusion matrix is a table showing true positives, true negatives, false positives, and false negatives, used to evaluate classification models.

Theory — Learn This for the Exam

Confusion matrix for binary classification:

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Derived metrics:

- $\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$
- $\text{Precision} = TP / (TP + FP)$
- $\text{Recall} = TP / (TP + FN)$
- $\text{Specificity} = TN / (TN + FP)$
- $F1 = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$

Question

What does the AUC-ROC curve measure?

- ☐ The accuracy of a regression model
- ☒ The ability of a classifier to distinguish between classes across all thresholds
- ☐ The speed of a clustering algorithm
- ☐ The number of clusters in a dataset

Correct Answer: B

Explanation

AUC-ROC (Area Under the Receiver Operating Characteristic Curve) measures how well a classifier distinguishes between classes across all possible thresholds.

Theory — Learn This for the Exam

ROC curve = True Positive Rate vs. False Positive Rate.

- X-axis: False Positive Rate (1 - Specificity)
- Y-axis: True Positive Rate (Sensitivity/Recall)
- Each point = a different classification threshold

AUC (Area Under Curve):

- Range: 0 to 1
- 0.5 = random classifier (no discrimination)
- 1.0 = perfect classifier
- Higher AUC = better discrimination

Exam tip: AUC measures discrimination ability, not accuracy. 0.5 is useless, 1.0 is perfect.

Question

What is the difference between precision and recall?

- ☐ Precision measures true positives out of all predicted positives; recall measures true positives out of all actual positives
- ☐ Precision measures true positives out of all actual positives; recall measures true positives out of all predicted positives
- ☐ Precision and recall are the same
- ☐ Precision is for regression; recall is for classification

Correct Answer: A

Explanation

Precision = $TP / (TP + FP)$; Recall = $TP / (TP + FN)$. Precision focuses on the accuracy of positive predictions; recall focuses on capturing all positive instances.

Module 2 — Theory: Precision vs. Recall

Theory — Learn This for the Exam

Precision: Of all predicted positives, how many are actually positive?

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall (Sensitivity): Of all actual positives, how many did we catch?

$$\text{Recall} = \frac{TP}{TP + FN}$$

When to use which:

- **High precision:** when false positives are costly (spam filter)
- **High recall:** when false negatives are costly (cancer detection, fraud detection)
- **F1 score:** harmonic mean of precision and recall

Exam tip: “Catch all positives” = recall. “Avoid false alarms” = precision.

Module 3 — Question 41

Question

Consider a medical diagnostic algorithm to diagnose a rare disease that occurs in 0.1% of the population. The disease is easily cured with a drug that has few side effects. Failing to treat the disease, however, will almost certainly lead to death. Which measure would you use to assess the algorithm's performance?

- ☒ Sensitivity
- ☐ Precision
- ☐ Accuracy
- ☐ Specificity

Correct Answer: A

Explanation

Sensitivity (true positive rate) is the correct measure because failing to detect the disease is fatal. We want to correctly identify all positive cases.

Module 3 — Theory: Sensitivity vs. Specificity

Theory — Learn This for the Exam

Sensitivity (Recall, True Positive Rate):

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

- Ability to correctly identify positive cases
- Use when false negatives are costly (missing a disease)

Specificity (True Negative Rate):

$$\text{Specificity} = \frac{TN}{TN + FP}$$

- Ability to correctly identify negative cases
- Use when false positives are costly (unnecessary treatment)

Exam tip: Rare disease + fatal if missed = sensitivity. Common disease + costly treatment = specificity.

Module 3 — Question 42

Question

A bank has implemented a model to detect fraudulent transactions. The model uses a data set with 10,000 randomly sampled transactions, 20 of which are fraudulent. What form of algorithmic bias is most likely to occur?

- ☐ Measurement bias
- ☐ Representation bias
- ☒ Data composition bias
- ☐ Historical bias

Correct Answer: C

Explanation

The dataset is highly imbalanced (20 fraudulent out of 10,000). When the dataset is not balanced, the model will underperform in identifying the minority group — this is data composition bias.

Theory — Learn This for the Exam

Types of bias:

- **Measurement bias:** features are measured differently across groups
- **Representation bias:** sample does not represent the population
- **Data composition bias:** dataset is imbalanced (minority class underrepresented)
- **Historical bias:** past discrimination is encoded in data
- **Problem specification bias:** target variable does not capture the real objective
- **Deployment bias:** model used in a context it was not designed for

Exam tip: Imbalanced dataset = data composition bias. Underrepresented group = representation bias.

Module 3 — Question 43

Question

A logistic model determines job candidate suitability based on work experience, certifications, and education level. To ensure education level does not affect predictions unfairly, what is the best approach?

- ☐ Remove the education level variable from the model
- ☒ Replace the education level with its average during prediction
- ☐ Incorporate interaction terms during training
- ☐ Create dummy variables to represent education level

Correct Answer: B

Explanation

Using the average education level during prediction ensures that people who are alike in all relevant respects but have different education levels are treated the same. Removing the variable does not fully eliminate its influence.

Theory — Learn This for the Exam

Fairness approaches:

- **Pre-processing:** fix the data before training
- **In-processing:** modify the algorithm during training
- **Post-processing:** adjust predictions after training

Bias mitigation techniques:

- Remove sensitive features (can leave proxy effects)
- Replace with average during prediction (equal treatment)
- Use fairness constraints during training
- Reweight samples to balance groups

Exam tip: Removing a variable does not remove its influence (proxies remain). Replacing with average ensures equal treatment.

Module 3 — Question 44

Question

A company is testing the fairness of a model that predicts purchasing decisions. How does demographic parity contribute to group fairness?

- ☐ Demographic parity ensures equal accuracy of purchasing rates across subgroups
- ☒ Demographic parity ensures identical predicted purchasing rates across subgroups
- ☐ Demographic parity ensures identical sensitivity of purchasing rates across subgroups
- ☐ Demographic parity ensures equal precision of purchasing rates across subgroups

Correct Answer: B

Explanation

Demographic parity requires that the predicted positive rate is the same across subgroups. It does not ensure equal accuracy, precision, or sensitivity.

Theory — Learn This for the Exam

Group fairness metrics:

- **Demographic parity:** equal positive prediction rates across groups
- **Equal opportunity:** equal true positive rates across groups
- **Equalized odds:** equal true positive and false positive rates
- **Predictive parity:** equal precision across groups

Individual fairness: similar individuals receive similar predictions. **Exam tip:** Demographic parity = equal prediction rates. Equal opportunity = equal recall.

Module 3 — Question 45

Question

A bank uses a chatbot for customer inquiries. Tests revealed incorrect information was provided. What is the most appropriate action?

- ☐ Upgrade the chatbot model by collecting more information
- ☐ Develop an ethical framework and review chatbot responses
- ☐ Generate periodic reports of risk analysis
- ☐ Implement design mechanisms that require human verification

Correct Answer: D

Explanation

Implementing design mechanisms that require periodic human input and verification mitigates the risk of incorrect information being provided to customers.

Theory — Learn This for the Exam

Human-in-the-loop (HITL) = human verification of AI outputs.

- Critical for high-stakes decisions
- Mitigates risks from AI errors
- Required by many regulations

Applications:

- Chatbots: human review of responses
- Medical diagnosis: doctor confirms AI recommendation
- Credit decisions: loan officer reviews AI decision
- Fraud detection: analyst investigates AI alerts

Exam tip: When AI makes errors with real-world consequences, human verification is the answer.

Question

What is the difference between model risk and algorithmic bias?

- ☐ Model risk is about model errors; algorithmic bias is about unfair discrimination
- ☐ Model risk is about unfair discrimination; algorithmic bias is about model errors
- ☐ They are the same thing
- ☐ Model risk applies to AI; algorithmic bias applies to statistics

Correct Answer: A

Explanation

Model risk refers to the potential for adverse consequences from model errors or misuse. Algorithmic bias refers to systematic and unfair discrimination in model outputs.

Theory — Learn This for the Exam

Model risk:

- Potential for adverse consequences from model errors
- Sources: wrong assumptions, bad data, implementation errors, misuse
- Applies to all models (statistical, ML, AI)

Algorithmic bias:

- Systematic and unfair discrimination in model outputs
- Sources: biased data, biased features, biased objectives
- Affects protected groups (race, gender, age)

Relationship: Bias is a source of model risk. **Exam tip:** Model risk = errors. Algorithmic bias = unfair discrimination.

Question

What is the purpose of explainability in AI models?

- ☐ To increase model accuracy
- ☒ To provide human-understandable reasons for model decisions
- ☐ To reduce training time
- ☐ To increase the number of features

Correct Answer: B

Explanation

Explainability aims to provide human-understandable explanations for how a model arrives at its decisions, which is important for trust, accountability, and regulatory compliance.

Module 3 — Theory: Explainability

Theory — Learn This for the Exam

Explainability = ability to explain model decisions in human terms.

- Why did the model make this prediction?
- Which features were most important?
- How would the prediction change if inputs changed?

Techniques:

- **SHAP values:** marginal contribution of each feature
- **LIME:** local approximation of model behavior
- **Feature importance:** global ranking of features
- **Partial dependence plots:** effect of a feature on predictions

Why it matters:

- Regulatory compliance (GDPR “right to explanation”)
- Trust and accountability
- Model validation and debugging

Module 3 — Question 48

Question

What is the difference between interpretability and explainability?

- ☐ Interpretability is about understanding the model's internal mechanisms; explainability is about explaining its decisions
- ☐ Interpretability is about explaining decisions; explainability is about understanding internal mechanisms
- ☐ They are the same thing
- ☐ Interpretability applies to linear models; explainability applies to neural networks

Correct Answer: A

Explanation

Interpretability refers to the ability to understand the internal workings of a model. Explainability refers to the ability to explain the model's decisions in human terms.

Module 3 — Theory: Interpretability vs. Explainability

Theory — Learn This for the Exam

Interpretability:

- Can you understand the model's internal logic?
- Linear regression: highly interpretable (coefficients)
- Deep neural network: not interpretable (black box)

Explainability:

- Can you explain a specific decision?
- Even black-box models can be explained post-hoc
- SHAP, LIME provide explanations

Relationship:

- Interpretable models are inherently explainable
- Non-interpretable models need post-hoc explainability tools

Exam tip: Interpretability = internal. Explainability = external explanation.

Question

What is the “black box” problem in AI?

- ☐ AI models are stored in black boxes
- ☒ AI models make decisions that cannot be understood by humans
- ☐ AI models are too large to fit in memory
- ☐ AI models are trained on black market data

Correct Answer: B

Explanation

The “black box” problem refers to AI models (especially deep neural networks) making decisions that cannot be easily understood or explained by humans.

Module 3 — Theory: The Black Box Problem

Theory — Learn This for the Exam

Black box = model whose internal workings are opaque.

- Deep neural networks have millions of parameters
- No simple formula explains decisions
- Cannot trace how inputs lead to outputs

Consequences:

- Difficult to validate
- Difficult to audit
- Difficult to trust
- Regulatory challenges

Solutions:

- Use interpretable models when possible
- Apply post-hoc explainability (SHAP, LIME)
- Use attention mechanisms (for NLP)

Question

What is the primary concern with using AI in hiring decisions?

- ☐ A AI is too slow
- ☒ B AI may perpetuate historical bias and discrimination
- ☐ C AI is too expensive
- ☐ D AI cannot process resumes

Correct Answer: B

Explanation

AI models trained on historical hiring data may perpetuate past discrimination, leading to unfair outcomes for protected groups.

Theory — Learn This for the Exam

Risks of AI in hiring:

- Historical bias: past discrimination encoded in data
- Proxy discrimination: features correlated with protected attributes
- Lack of transparency: candidates cannot understand decisions
- Regulatory risk: violation of anti-discrimination laws

Mitigation:

- Audit training data for bias
- Remove or de-bias proxy features
- Use fairness constraints
- Provide explanations to candidates
- Human review of AI decisions

Exam tip: AI in hiring is a classic example of algorithmic bias risk.

Module 4 — Question 51

Question

A sport analyst is developing an AI system to predict winners of a basketball league. The analyst would like to explain the model by understanding the marginal effect of each feature. Which technique is best?

- ☒ Shapley values
- ☐ Feature importance scores
- ☐ Grid search
- ☐ Feature scaling

Correct Answer: A

Explanation

Shapley values provide the marginal contribution of each feature to the prediction, making them ideal for understanding marginal effects.

Module 4 — Theory: Shapley Values

Theory — Learn This for the Exam

Shapley values = marginal contribution of each feature.

- From cooperative game theory
- Fairly distributes the “payout” (prediction) among features
- Considers all possible feature subsets

Properties:

- Additivity: sum of Shapley values = prediction - baseline
- Consistency: if a feature's contribution increases, its Shapley value increases
- Symmetry: identical features get identical values

vs. Feature importance:

- Feature importance: global ranking (not per-prediction)
- Shapley values: per-prediction attribution

Exam tip: Marginal effect of each feature = Shapley values.

Module 4 — Question 52

Question

A bank uses a model to determine if loan applications should be approved. Customers have raised concerns about how the model reaches the decision. Which is the source of reputational risk and what can be done?

- ☐ Privacy Breaches — Mitigated by securing data access
- ☐ Algorithmic Bias — Mitigated by using diverse training data
- ☒ Lack of Explainability — Mitigated by providing reasons for a loan being rejected
- ☐ Lack of Transparency — Mitigated by disclosing the full details of the training data

Correct Answer: C

Explanation

Customers want to understand why their loan was rejected. Lack of explainability is the source of reputational risk. Providing reasons for rejection mitigates this.

Theory — Learn This for the Exam

Sources of AI-related reputational risk:

- **Lack of explainability:** customers do not understand decisions
- **Algorithmic bias:** unfair treatment of groups
- **Privacy breaches:** mishandling personal data
- **Lack of transparency:** hiding how AI works

Mitigation:

- **Explainability:** provide reasons for decisions
- **Bias audits:** test for fairness
- **Privacy:** secure data, get consent
- **Transparency:** disclose AI use, provide recourse

Exam tip: “Why was I rejected?” = lack of explainability.

Question

Which of the following is considered an advantage of implementing a practical ethics framework?

- ☒ A practical ethics framework can provide clear guidelines for evaluating ethical dilemmas
- ☐ A practical ethics framework can eliminate conflicts of interest between stakeholders
- ☐ A practical ethics framework can guarantee compliance with legal requirements
- ☐ A practical ethics framework can simplify decision making by focusing on profit maximization

Correct Answer: A

Explanation

A practical ethics framework provides clear guidelines for evaluating ethical dilemmas. It cannot eliminate conflicts of interest, guarantee legal compliance, or focus only on profit.

Module 4 — Theory: Practical Ethics Frameworks

Theory — Learn This for the Exam

Practical ethics framework = structured approach to ethical decisions.

- Provides guidelines for evaluating dilemmas
- Helps ensure consistency
- Supports accountability
- Builds trust with stakeholders

What it does NOT do:

- Eliminate all conflicts of interest
- Guarantee legal compliance
- Focus only on profit
- Replace human judgment

Components:

- Principles (beneficence, nonmaleficence, justice, autonomy)
- Processes (review, escalation, documentation)

Question

What is the most likely advantage of implementing an AI ethics framework?

- ☐ An AI ethics framework helps increase the accuracy of the models it is applied to
- ☐ An AI ethics framework strengthens the firewall that defends against hackers
- ☐ An AI ethics framework helps increase the productivity of model testing
- ☒ An AI ethics framework helps reduce the risk of regulatory non-compliance

Correct Answer: D

Explanation

An AI ethics framework helps reduce the risk of regulatory non-compliance by ensuring AI systems align with legal and ethical standards.

Theory — Learn This for the Exam

Why AI ethics frameworks matter:

- Regulatory compliance (EU AI Act, GDPR, CCPA)
- Reputational protection
- Stakeholder trust
- Risk management

Key regulations:

- **EU AI Act:** risk-based classification of AI systems
- **GDPR:** right to explanation, data protection
- **CCPA:** California privacy rights
- **SEC/FCA:** model risk management guidance

Exam tip: Ethics frameworks reduce regulatory risk, not model accuracy or cybersecurity.

Question

Utilitarianism is one ethical theory that can be incorporated into an AI ethics framework. Which statement is correct regarding Utilitarianism?

- ☒ Utilitarian approach can lead to violations of individual rights
- ☐ Utilitarian approach tends to focus on virtuous character traits
- ☐ Utilitarian approach relies heavily on laws and regulations
- ☐ Utilitarian approach values the intrinsic morality of actions

Correct Answer: A

Explanation

Utilitarianism focuses on maximizing overall good, which can lead to actions that violate individual rights for the greater good.

Theory — Learn This for the Exam

Three main ethical frameworks:

- **Consequentialism (Utilitarianism):**

- Judges actions by their consequences
- Goal: maximize overall good/happiness
- Can justify violating individual rights for greater good

- **Deontology:**

- Judges actions by adherence to rules and duties
- Some actions are inherently right or wrong
- Does not consider consequences

- **Virtue Ethics:**

- Focuses on character traits and virtues
- What would a virtuous person do?
- Emphasis on living a good life

Exam tip: Utilitarianism = consequences. Deontology = rules. Virtue = character.

Question

The CRO of a large international bank is considering ethical frameworks for AI models. A consultant proposes a deontological ethics framework. Which best describes a characteristic of deontology?

- ☐ A deontological ethical framework judges morality based on consequences
- ☒ A deontological ethical framework judges morality based on adherence to ethical duties and rules
- ☐ A deontological ethical framework provides a single, quantifiable metric for moral value
- ☐ A deontological ethical framework emphasizes virtuous character traits

Correct Answer: B

Explanation

Deontology judges morality based on adherence to ethical duties and rules, not on consequences.

Theory — Learn This for the Exam

Deontology = duty-based ethics.

- Actions are right or wrong in themselves
- Consequences do not determine morality
- Key concept: categorical imperative (Kant)
- “Do not lie” is a duty, regardless of consequences

vs. Consequentialism:

- Consequentialism: the ends justify the means
- Deontology: the means matter, not just the ends

Application to AI:

- Certain AI uses are inherently wrong (e.g., mass surveillance)
- Rules must be followed regardless of benefits

Exam tip: Deontology = rules and duties. Consequentialism = outcomes.

Module 4 — Question 57

Question

A government legislator is considering the principle of justice in AI ethics. Which is correct regarding justice?

- ☐ Conflicts can exist between different concepts of distributive justice such as egalitarianism and utilitarianism
- ☐ Impartial application of the law is not typically considered as a requirement to achieve procedural justice
- ☐ Personal autonomy is generally considered secondary to achieving social justice
- ☐ Meritocracy is not one of the principles of distributive justice

Correct Answer: A

Explanation

Different concepts of justice (egalitarianism, utilitarianism, meritocracy) can conflict with each other. Procedural justice requires impartial application of the law.

Theory — Learn This for the Exam

Types of justice:

- **Distributive justice:** fair allocation of resources
 - Egalitarianism: equal distribution
 - Utilitarianism: maximize total welfare
 - Meritocracy: distribute based on merit
- **Procedural justice:** fair processes
 - Impartial application of the law
 - Transparent decision-making
- **Social justice:** fairness in society

Conflicts: Different concepts of justice can conflict (e.g., egalitarianism vs. meritocracy). **Exam tip:** Justice concepts can conflict. Procedural justice requires impartiality.

Module 4 — Question 58

Question

An AI system recommends treatment plans for unconscious patients. In some cases, the AI suggests a treatment that conflicts with the patient's personal preferences. Which AI ethics principle is violated?

- ☐ Explainability
- ☐ Beneficence
- ☐ Autonomy
- ☐ Non-maleficence

Correct Answer: C

Explanation

Autonomy refers to respecting the patient's right to make their own decisions. The AI conflicting with patient preferences violates autonomy.

Theory — Learn This for the Exam

Five principles of AI ethics:

- **Nonmaleficence:** do no harm
- **Beneficence:** do good
- **Justice:** fairness
- **Autonomy:** respect individual choice
- **Explainability:** provide understandable reasons

Autonomy in AI:

- Patients have the right to refuse treatment
- Users must consent to AI decisions
- AI should support, not replace, human judgment

Exam tip: “Conflicts with patient preferences” = autonomy violation.

Module 4 — Question 59

Question

A pharmaceutical company developed a model for tropical diseases using data from travelers returning to the US and Canada. The model performed poorly in Brazil, Nigeria, and Thailand. What is the most likely cause?

- ☐ Bias in problem specification
- ☐ Bias in model deployment
- ☐ Historical bias
- ☒ Sampling bias

Correct Answer: D

Explanation

College students from the US and Canada may not be representative of hospital patients in Brazil, Nigeria, and Thailand. This is sampling bias.

Module 4 — Theory: Sampling Bias

Theory — Learn This for the Exam

Sampling bias = sample does not represent the target population.

- Training data from one population
- Deployment in a different population
- Model performs poorly on new population

Examples:

- Medical AI trained on US travelers, deployed in developing countries
- Hiring AI trained on one demographic, deployed globally
- Credit model trained on one region, deployed in another

Mitigation:

- Collect representative data
- Test on diverse populations
- Monitor performance across subgroups
- Use transfer learning

Question

What is the primary purpose of an AI ethics framework?

- ☐ To maximize model accuracy
- ☒ To provide guidelines for ethical AI development and use
- ☐ To reduce training time
- ☐ To increase the number of features

Correct Answer: B

Explanation

An AI ethics framework provides guidelines for ethical AI development and use, ensuring AI systems align with societal values and legal requirements.

Theory — Learn This for the Exam

AI ethics framework = structured approach to ethical AI.

- Defines principles (beneficence, nonmaleficence, justice, autonomy, explainability)
- Establishes processes (review, escalation, documentation)
- Assigns roles (ethics committee, AI ethics officer)
- Provides training and awareness

Benefits:

- Reduces regulatory risk
- Builds trust
- Ensures consistency
- Protects reputation

Exam tip: Ethics framework = guidelines for ethical AI, not technical accuracy.

Module 5 — Question 61

Question

The CRO of an investment firm oversees model risk governance. What is a crucial consideration when evaluating the model landscape?

- ☐ Verify that each model operates independently
- ☐ Ensure that all models use the same input data
- ☐ Assess whether models can be consolidated
- ☒ Ensure that interdependencies between models are identified

Correct Answer: D

Explanation

A model landscape illustrates the interactions among models. Identifying interdependencies is crucial to understanding how one model's failure could affect others.

Theory — Learn This for the Exam

Model landscape = overview of all models and their interactions.

- Shows relationships between models
- Identifies critical models
- Maps data flows
- Highlights interdependencies

Why interdependencies matter:

- One model's output may be another's input
- Failure can cascade
- Correlated errors can amplify risk
- Systemic risk across the organization

Exam tip: Model landscape = map of models and their interdependencies.

Question

A sovereign wealth fund is reviewing model governance at a bank. Which correctly describes roles and responsibilities?

- ☐ The board of directors should maintain a model inventory
- ☐ The model risk function should set the model-risk appetite
- ☐ Model ownership should never be shared
- ☐ Internal and external auditors are necessary independent checks on model validation

Correct Answer: D

Explanation

Internal and external auditors provide independent checks on the model validation process. The board sets risk appetite, and the model risk function maintains the inventory.

Theory — Learn This for the Exam

Key roles in model governance:

- **Board of directors:** sets model risk appetite
- **Model risk function:** maintains model inventory, develops policies
- **Model developers:** build and document models
- **Model validators:** independently validate models
- **Internal auditors:** independent check on validation
- **External auditors:** external independent check
- **Model owners:** accountable for model use

Exam tip:

- Board sets appetite
- Model risk function maintains inventory
- Auditors check validation

Question

When implementing a new ML model, which team ensures integration with existing IT systems does not disrupt operations?

-  The Core Implementation Team
-  The System Implementation Team
-  The Model Design Team
-  The Data Management Team

Correct Answer: B

Explanation

The system implementation team is responsible for ensuring the model is properly integrated with existing IT systems and functions correctly in the environment.

Theory — Learn This for the Exam

Teams involved in model implementation:

- **Model Design Team:** designs the model architecture and methodology
- **Core Implementation Team:** builds and codes the model
- **System Implementation Team:** integrates model with IT systems
- **Data Management Team:** ensures data quality and availability

System implementation team responsibilities:

- Integration with existing IT infrastructure
- Ensuring model functions correctly in production
- Managing deployment
- Monitoring system performance

Exam tip: IT integration = System Implementation Team.

Module 5 — Question 64

Question

A bank developed a new AI model for mortgage prepayments. It outperformed the traditional model. What is the appropriate next step?

- ☐ Deployment of the model for production use
- ☐ Informing IT that a new model needs to be rolled out
- ☐ Retirement of the existing model
- ☒ Requesting an initial validation of the model from the model validation team

Correct Answer: D

Explanation

Before deployment, the model must be independently validated by the model validation team. This is a regulatory requirement and best practice.

Theory — Learn This for the Exam

Model validation = independent assessment of model.

- Conducted by independent team (not developers)
- Assesses conceptual soundness
- Evaluates data quality
- Tests model performance
- Reviews implementation

When to validate:

- Before initial deployment
- Periodically (annual or risk-based)
- After significant changes
- When performance degrades

Exam tip: New model → validate before deployment. Never deploy without validation.

Question

A bank developed a new AI credit scoring model using alternative data (school transcripts, social media, landlord-tenant disputes). What is an appropriate finding by the validation team?

- ☐ No material concerns, as the augmented data appears appropriate
- ☒ Raise material concerns around potentially improper inclusion of sensitive alternative data
- ☐ Approve the model if back testing showed improvement
- ☐ Recommend that additional alternative data be included

Correct Answer: B

Explanation

Use of alternative data requires heightened scrutiny. Sensitive data used to make credit decisions could give rise to ethical breaches and may not have been approved by the applicant.

Theory — Learn This for the Exam

Alternative data = non-traditional data sources.

- Social media, school transcripts, rent payments
- Can expand credit access
- Raises ethical and legal concerns

Data governance requirements:

- **Provenance:** legal right to use data
- **Consent:** applicants must approve use
- **Privacy:** protect sensitive information
- **Fairness:** avoid discriminatory features
- **Transparency:** disclose data use

Exam tip: Alternative data = heightened scrutiny. Sensitive data = material concerns.

Module 5 — Question 66

Question

In a financial institution, ML models are exposed to rapidly changing data environments. What is an advantage of increasing the frequency of model monitoring?

- ☐ It reduces the need for model retraining
- ☒ It allows detection of performance degradation or drift, enabling timely intervention
- ☐ It minimizes the involvement of the IT team
- ☐ It improves the model's interpretability

Correct Answer: B

Explanation

Increasing monitoring frequency allows early detection of performance degradation or drift, enabling timely intervention to maintain model accuracy.

Theory — Learn This for the Exam

Model monitoring = ongoing oversight of model performance.

- Detect performance degradation
- Identify data drift
- Identify concept drift
- Trigger retraining or recalibration

Types of drift:

- **Data drift:** input distribution changes
- **Concept drift:** relationship between inputs and outputs changes
- **Label drift:** target distribution changes

Monitoring frequency:

- Higher risk = more frequent monitoring
- Rapidly changing environments = more frequent
- Stable environments = less frequent

Question

What is the purpose of a model inventory?

- ☐ To store model code
- ☒ To list all models in use and track their risk
- ☐ To train models
- ☐ To visualize model performance

Correct Answer: B

Explanation

A model inventory is a comprehensive list of all models in use, tracking their purpose, risk level, owner, and validation status.

Theory — Learn This for the Exam

Model inventory = registry of all models.

- Maintained by model risk function
- Tracks all models in use
- Documents purpose, owner, risk level
- Records validation status
- Identifies interdependencies

Contents:

- Model name and description
- Purpose and use case
- Owner and developer
- Risk classification
- Validation date and status
- Last review date

Question

What is the primary purpose of model documentation?

- ☐ To increase model accuracy
- ☒ To provide a record of model design, assumptions, and limitations
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model documentation provides a record of model design, assumptions, limitations, and usage, supporting validation, audit, and governance.

Theory — Learn This for the Exam

Model documentation = comprehensive record of the model.

- Purpose and intended use
- Methodology and assumptions
- Data sources and preparation
- Model limitations
- Performance metrics
- Validation results
- Usage guidelines

Why it matters:

- Supports validation
- Enables audit
- Facilitates knowledge transfer
- Ensures regulatory compliance

Question

What is the purpose of the use test in model validation?

- ☐ To test if the model is used in production
- ☒ To assess whether the model is appropriate for its intended use
- ☐ To test the model's speed
- ☐ To test the model's accuracy

Correct Answer: B

Explanation

The use test assesses whether the model is appropriate for its intended use and whether users understand its limitations.

Theory — Learn This for the Exam

Use test = assess whether model is fit for its intended purpose.

- Is the model being used as intended?
- Do users understand its limitations?
- Is the model appropriate for the business context?
- Are outputs being interpreted correctly?

Questions to ask:

- What decisions does the model support?
- Who uses the model?
- Do users understand the model's assumptions?
- Are there alternative models that would be better?

Exam tip: Use test = is the model appropriate for its intended use?

Question

What is the difference between white-box and black-box testing?

- ☐ White-box tests internal logic; black-box tests inputs and outputs
- ☐ White-box tests inputs and outputs; black-box tests internal logic
- ☐ They are the same
- ☐ White-box is for AI; black-box is for statistics

Correct Answer: A

Explanation

White-box testing examines the internal logic and code of the model. Black-box testing examines inputs and outputs without knowledge of internal workings.

Theory — Learn This for the Exam

White-box testing:

- Examines internal logic and code
- Requires knowledge of model structure
- Tests specific components
- Identifies implementation errors

Black-box testing:

- Examines inputs and outputs only
- No knowledge of internal workings
- Tests overall behavior
- Simulates real-world use

Both are needed for comprehensive validation. Exam tip: White-box = internal. Black-box = external.

Question

What is the purpose of data provenance in AI governance?

- ☒ To verify the legal right to use data
- ☐ To increase data volume
- ☐ To speed up training
- ☐ To reduce data quality

Correct Answer: A

Explanation

Data provenance verifies the legal right to use data, especially alternative data, ensuring compliance with regulations and ethical standards.

Theory — Learn This for the Exam

Data provenance = origin and history of data.

- Where did the data come from?
- Do we have the legal right to use it?
- Was consent obtained?
- Is it accurate and up-to-date?

Why it matters:

- Legal compliance (GDPR, CCPA)
- Ethical use of data
- Regulatory requirements
- Reputational protection

Exam tip: Provenance = legal right to use data.

Question

What is the primary purpose of data classification?

- ☐ To increase data volume
- ☒ To categorize data by sensitivity and structure
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Data classification categorizes data by sensitivity (public, internal, confidential, restricted) and structure (structured, unstructured) to determine appropriate handling.

Theory — Learn This for the Exam

Data classification = categorizing data by sensitivity and structure.

- **Sensitivity levels:**

- Public: freely available
- Internal: for employees only
- Confidential: restricted access
- Restricted: highly sensitive (PII, PHI)

- **Structure:**

- Structured: tables, databases
- Unstructured: text, images, audio

Purpose: Determine appropriate handling, access controls, and protection. **Exam tip:** Classification = sensitivity and structure.

Question

What is the purpose of metadata management?

- ☐ To increase data volume
- ☒ To document data origin, structure, definitions, and relationships
- ☐ To speed up training
- ☐ To reduce data quality

Correct Answer: B

Explanation

Metadata management documents data origin, structure, definitions, and relationships to other data, supporting data governance and usability.

Theory — Learn This for the Exam

Metadata = data about data.

- Origin and history
- Structure and format
- Definitions and business meaning
- Relationships to other data
- Access permissions

Why it matters:

- Data discovery and understanding
- Data lineage tracking
- Regulatory compliance
- Effective data governance

Exam tip: Metadata = data about data (origin, structure, definitions, relationships).

Question

What is the purpose of data access controls?

- ☐ To increase data volume
- ☒ To ensure only authorized users can access data
- ☐ To speed up training
- ☐ To reduce data quality

Correct Answer: B

Explanation

Data access controls ensure only authorized users can access data, protecting privacy and security.

Theory — Learn This for the Exam

Data access controls = who can access what data.

- Role-based access control (RBAC)
- Attribute-based access control (ABAC)
- Least privilege principle
- Need-to-know basis

Why it matters:

- Privacy protection
- Security
- Regulatory compliance
- Preventing misuse

Exam tip: Access controls = authorized users only.

Question

What is the primary purpose of a data governance framework?

- ☐ To increase data volume
- ☒ To ensure data quality, security, and compliance
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

A data governance framework ensures data quality, security, compliance, and proper use across the organization.

Theory — Learn This for the Exam

Data governance = framework for managing data as an asset.

- Data strategy and vision
- Data quality (accuracy, consistency, integrity)
- Data provenance and legal rights
- Data classification
- Metadata management
- Data protection and security
- Data access controls
- Compliance with regulations
- Roles and responsibilities

Exam tip: Data governance = quality, security, compliance, proper use.

Question

What is the purpose of model risk management?

- ☐ To increase model accuracy
- ☒ To identify, assess, and mitigate risks from models
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model risk management identifies, assesses, and mitigates risks from model errors, misuse, and limitations.

Theory — Learn This for the Exam

Model risk management = framework for managing model risk.

- Model identification and inventory
- Risk assessment and classification
- Model development and documentation
- Independent validation
- Ongoing monitoring
- Governance and oversight

Sources of model risk:

- Wrong assumptions
- Bad data
- Implementation errors
- Misuse
- Model drift

Question

What is the role of the model risk function?

- ☐ To build models
- ☒ To maintain the model inventory and develop policies
- ☐ To use models
- ☐ To reduce the number of features

Correct Answer: B

Explanation

The model risk function maintains the model inventory, develops policies, and oversees model risk management across the organization.

Theory — Learn This for the Exam

Model risk function responsibilities:

- Maintain model inventory
- Develop model risk policies
- Classify model risk
- Coordinate validation
- Monitor model performance
- Report to senior management
- Ensure regulatory compliance

Independence: Must be independent from model developers. **Exam tip:** Model risk function = inventory + policies.

Question

What is the purpose of periodic model validation?

- ☐ To increase model accuracy
- ☒ To confirm the model remains fit for purpose over time
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Periodic model validation confirms the model remains aligned with its intended purpose, assumptions remain valid, and performance is as expected.

Theory — Learn This for the Exam

Periodic validation = ongoing assessment of model fitness.

- Confirm assumptions remain valid
- Verify data is still appropriate
- Check performance hasn't degraded
- Ensure methodology reflects best practices
- Confirm model still aligns with purpose

Frequency:

- Based on model risk ranking
- Higher risk = more frequent
- Typically annually or risk-based
- Also triggered by significant changes

Exam tip: Periodic validation = is the model still fit for purpose?

Question

What is the purpose of model implementation tasks?

- ☐ To design the model
- ☒ To deploy the model into production and integrate with IT systems
- ☐ To validate the model
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model implementation tasks involve deploying the model into production, integrating with IT systems, and ensuring it functions correctly in the environment.

Theory — Learn This for the Exam

Model implementation = deploying model into production.

- Integration with IT systems
- Data pipeline setup
- Performance testing
- User training
- Documentation
- Monitoring setup

Challenges:

- Legacy system integration
- Data quality issues
- Scalability
- Security
- Change management

Question

What is the purpose of model adaptation tasks?

- ☐ To design the model
- ☒ To update the model to reflect changing conditions
- ☐ To validate the model
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model adaptation tasks update the model to reflect changing conditions, such as new data, changing business needs, or regulatory requirements.

Theory — Learn This for the Exam

Model adaptation = updating model for changing conditions.

- Retraining with new data
- Recalibrating parameters
- Updating assumptions
- Adding new features
- Responding to regulatory changes

Triggers:

- Performance degradation
- Data drift
- Concept drift
- Business changes
- Regulatory changes

Exam tip: Adaptation = updating model for changing conditions.

Question

What is the purpose of a model risk appetite?

- ☐ To increase model accuracy
- ☒ To define the level of model risk the organization is willing to accept
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model risk appetite defines the level of model risk the organization is willing to accept, set by the board of directors.

Theory — Learn This for the Exam

Model risk appetite = level of model risk organization accepts.

- Set by board of directors
- Defines tolerance for model risk
- Guides model risk management
- Aligns with overall risk appetite

Considerations:

- Business strategy
- Regulatory requirements
- Risk capacity
- Stakeholder expectations

Exam tip: Board sets model risk appetite.

Question

What is the purpose of model governance policies?

- ☐ To increase model accuracy
- ☒ To establish rules and procedures for model development, validation, and use
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model governance policies establish rules and procedures for model development, validation, and use, ensuring consistency and accountability.

Theory — Learn This for the Exam

Model governance policies = rules for model management.

- Model development standards
- Validation requirements
- Documentation standards
- Approval processes
- Monitoring requirements
- Roles and responsibilities

Purpose:

- Consistency across models
- Accountability
- Regulatory compliance
- Risk management

Exam tip: Policies = rules for model management.

Question

What is the purpose of model documentation standards?

- ☐ To increase model accuracy
- ☒ To ensure consistent and comprehensive documentation of models
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model documentation standards ensure consistent and comprehensive documentation of models, supporting validation, audit, and knowledge transfer.

Theory — Learn This for the Exam

Documentation standards = requirements for model documentation.

- Purpose and intended use
- Methodology and assumptions
- Data sources and preparation
- Model limitations
- Performance metrics
- Validation results
- Usage guidelines

Benefits:

- Consistency
- Knowledge transfer
- Audit readiness
- Regulatory compliance

Question

What is the purpose of model inventory and model landscape?

- ☐ To increase model accuracy
- ☒ To track all models and their interdependencies
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model inventory tracks all models in use. Model landscape shows interactions and interdependencies between models.

Module 5 — Theory: Model Inventory vs. Model Landscape

Theory — Learn This for the Exam

Model inventory:

- List of all models
- Name, purpose, owner, risk level
- Validation status
- Last review date

Model landscape:

- Visual representation of model interactions
- Shows data flows between models
- Identifies critical models
- Highlights interdependencies

Exam tip: Inventory = list. Landscape = map of interactions.

Question

What is the purpose of model risk reporting?

- ☐ To increase model accuracy
- ☒ To inform senior management and board about model risk
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model risk reporting informs senior management and the board about model risk, including inventory, validation status, and key issues.

Theory — Learn This for the Exam

Model risk reporting = communication of model risk to stakeholders.

- Model inventory summary
- Validation status
- Key risk issues
- Performance metrics
- Remediation plans

Audience:

- Board of directors
- Senior management
- Risk committee
- Regulators

Exam tip: Reporting = informing management about model risk.

Question

What is the purpose of model change management?

- ☐ To increase model accuracy
- ☒ To control changes to models and ensure proper review
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model change management controls changes to models, ensuring proper review, approval, and documentation of changes.

Theory — Learn This for the Exam

Change management = controlling changes to models.

- Identify need for change
- Assess impact
- Obtain approval
- Implement change
- Document change
- Validate change

Types of changes:

- Minor: parameter updates
- Major: methodology changes
- Emergency: urgent fixes

Exam tip: Change management = control and review of model changes.

Question

What is the purpose of model review?

- ☐ To increase model accuracy
- ☒ To periodically assess model performance and continued fitness
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model review periodically assesses model performance and continued fitness, identifying issues and recommending actions.

Theory — Learn This for the Exam

Model review = periodic assessment of model.

- Performance monitoring
- Assumption validation
- Data quality check
- Methodology review
- Documentation update

Frequency:

- Based on model risk
- Higher risk = more frequent
- Triggered by significant changes

Outcomes:

- Continue use
- Modify model

Question

What is the purpose of model retirement?

- ☐ To increase model accuracy
- ☒ To formally decommission a model that is no longer needed
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model retirement formally decommissions a model that is no longer needed, ensuring proper documentation and transition.

Theory — Learn This for the Exam

Model retirement = formal decommissioning of a model.

- Document reason for retirement
- Obtain approval
- Transition to replacement model
- Archive documentation
- Update model inventory

Reasons for retirement:

- Model no longer needed
- Better model available
- Regulatory changes
- Business changes

Exam tip: Retirement = formal decommissioning.

Question

What is the purpose of model risk culture?

- ☐ To increase model accuracy
- ☒ To promote awareness and accountability for model risk across the organization
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model risk culture promotes awareness and accountability for model risk across the organization, from board to front-line staff.

Theory — Learn This for the Exam

Model risk culture = shared values and behaviors around model risk.

- Awareness of model risk
- Accountability at all levels
- Open communication
- Continuous learning
- Ethical behavior

Building blocks:

- Tone from the top
- Training and education
- Incentives aligned with risk management
- Reporting and transparency

Exam tip: Culture = awareness and accountability for model risk.

Question

What is the purpose of model risk training?

- ☐ To increase model accuracy
- ☒ To ensure staff understand model risk and their responsibilities
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model risk training ensures staff understand model risk and their responsibilities in managing it.

Theory — Learn This for the Exam

Model risk training = education on model risk.

- What is model risk?
- Roles and responsibilities
- Policies and procedures
- Best practices
- Case studies

Audience:

- Board and senior management
- Model developers
- Model validators
- Model users
- All staff

Exam tip: Training = education on model risk.

Question

A bank uses AI for fraud detection. The model is highly effective but the risk team is concerned about inscrutability. Which statement correctly describes inscrutability?

- ☒ Inscrutability is the inability to understand the internal mechanisms of a model
- ☐ Inscrutability is the inability to determine accuracy
- ☐ Inscrutability is the inability to explain predictions
- ☐ Inscrutability is the inability to generalize

Correct Answer: A

Explanation

Inscrutability refers to the inability to understand the internal mechanisms of a model — how it arrives at its outputs.

Theory — Learn This for the Exam

Inscrutability = model risk.

- Cannot understand internal workings
- Difficult to validate
- Difficult to audit
- Regulatory challenges

Mitigation:

- Use interpretable models when possible
- Apply post-hoc explainability
- Document model behavior
- Human oversight

Exam tip: Inscrutability = cannot understand internal mechanisms.

Question

What is the primary purpose of a data governance framework?

- ☐ To increase data volume
- ☒ To ensure data quality, security, and compliance
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

A data governance framework ensures data quality, security, and compliance across the organization.

Theory — Learn This for the Exam

Data governance = managing data as an asset.

- Data strategy and vision
- Data quality
- Data provenance
- Data classification
- Metadata management
- Data protection
- Data access
- Compliance
- Roles and responsibilities

Exam tip: Data governance = quality, security, compliance.

Question

What is the purpose of model risk appetite?

- ☐ To increase model accuracy
- ☒ To define the level of model risk the organization is willing to accept
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model risk appetite defines the level of model risk the organization is willing to accept, set by the board of directors.

Theory — Learn This for the Exam

Model risk appetite = level of model risk accepted.

- Set by board of directors
- Defines tolerance for model risk
- Guides model risk management
- Aligns with overall risk appetite

Exam tip: Board sets model risk appetite.

Question

What is the purpose of model validation?

- ☐ To increase model accuracy
- ☒ To independently assess model soundness and fitness
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model validation independently assesses model soundness, data quality, performance, and fitness for purpose.

Theory — Learn This for the Exam

Model validation = independent assessment.

- Conceptual soundness
- Data quality
- Performance testing
- Implementation review
- Documentation review

Independence: Must be conducted by team independent from developers. **When:** Before deployment, periodically, after changes. **Exam tip:** Validation = independent assessment of model.

Question

What is the purpose of model monitoring?

- ☐ To increase model accuracy
- ☒ To detect performance degradation and drift
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model monitoring detects performance degradation and drift, enabling timely intervention.

Theory — Learn This for the Exam

Model monitoring = ongoing oversight.

- Performance metrics
- Data drift detection
- Concept drift detection
- Alert thresholds
- Reporting

Triggers for action:

- Performance below threshold
- Significant drift
- Regulatory changes
- Business changes

Exam tip: Monitoring = detect degradation and drift.

Question

What is the purpose of model documentation?

- ☐ To increase model accuracy
- ☒ To provide a record of model design, assumptions, and limitations
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model documentation provides a record of model design, assumptions, limitations, and usage, supporting validation and audit.

Theory — Learn This for the Exam

Model documentation = comprehensive record.

- Purpose and intended use
- Methodology and assumptions
- Data sources
- Limitations
- Performance metrics
- Validation results
- Usage guidelines

Exam tip: Documentation = record of design, assumptions, limitations.

Question

What is the purpose of a model inventory?

- ☐ To increase model accuracy
- ☒ To list all models in use and track their risk
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

A model inventory lists all models in use, tracking their purpose, risk level, owner, and validation status.

Theory — Learn This for the Exam

Model inventory = registry of all models.

- Model name and description
- Purpose and use case
- Owner and developer
- Risk classification
- Validation status
- Last review date

Maintained by: Model risk function. **Exam tip:** Inventory = list of all models with risk and validation status.

Question

What is the purpose of model landscape?

- ☐ To increase model accuracy
- ☒ To show interactions and interdependencies between models
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model landscape shows interactions and interdependencies between models, identifying critical models and data flows.

Theory — Learn This for the Exam

Model landscape = map of model interactions.

- Shows relationships between models
- Identifies critical models
- Maps data flows
- Highlights interdependencies

Why it matters:

- One model's output may be another's input
- Failure can cascade
- Correlated errors amplify risk

Exam tip: Landscape = map of model interactions.

Question

What is the purpose of model risk reporting?

- ☐ To increase model accuracy
- ☒ To inform senior management and board about model risk
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model risk reporting informs senior management and the board about model risk, including inventory, validation status, and key issues.

Theory — Learn This for the Exam

Model risk reporting = communication of model risk.

- Model inventory summary
- Validation status
- Key risk issues
- Performance metrics
- Remediation plans

Audience: Board, senior management, risk committee, regulators. **Exam tip:** Reporting = informing management about model risk.

Question

What is the purpose of model change management?

- ☐ To increase model accuracy
- ☒ To control changes to models and ensure proper review
- ☐ To speed up training
- ☐ To reduce the number of features

Correct Answer: B

Explanation

Model change management controls changes to models, ensuring proper review, approval, and documentation.

Theory — Learn This for the Exam

Change management = controlling changes to models.

- Identify need for change
- Assess impact
- Obtain approval
- Implement change
- Document change
- Validate change

Types of changes:

- Minor: parameter updates
- Major: methodology changes
- Emergency: urgent fixes

Exam tip: Change management = control and review of model changes.

Final Review — Key Takeaways

Module 1: AI and Risk

- Inscrutability vs. explainability vs. interpretability
- Classical AI vs. deep learning
- Data types, scaling, PCA

Module 2: Tools & Techniques

- K-means, hierarchical clustering, DBSCAN
- WCSS, BCSS, silhouette, scree plot
- Random forest, SVM, NLP, regularization
- Classification metrics: precision, recall, F1, AUC

Module 3: Risk and Risk Factors

- Sensitivity vs. specificity
- Algorithmic bias sources

Good Luck!

You are ready for the GARP Risk and AI exam.

Unofficial study aid — not affiliated with GARP.